

Statistics for Social Science Research

Lecture 3

20 August 2026

Topics for today

- Variability (Chapter 4, Statistics for the Behavioral Sciences)
 - Standard deviation and variance for a population
 - Standard deviation and variance for samples
- Z-Scores (Chapter 5, Statistics for the Behavioral Sciences)

(The slides in this power point are partly taken from the Power Point Lecture Slides to accompany the textbook “Essentials of Statistics for the Behavioral Sciences” by Frederick J Gravetter and Larry B. Wallnau.)

Chapter 4

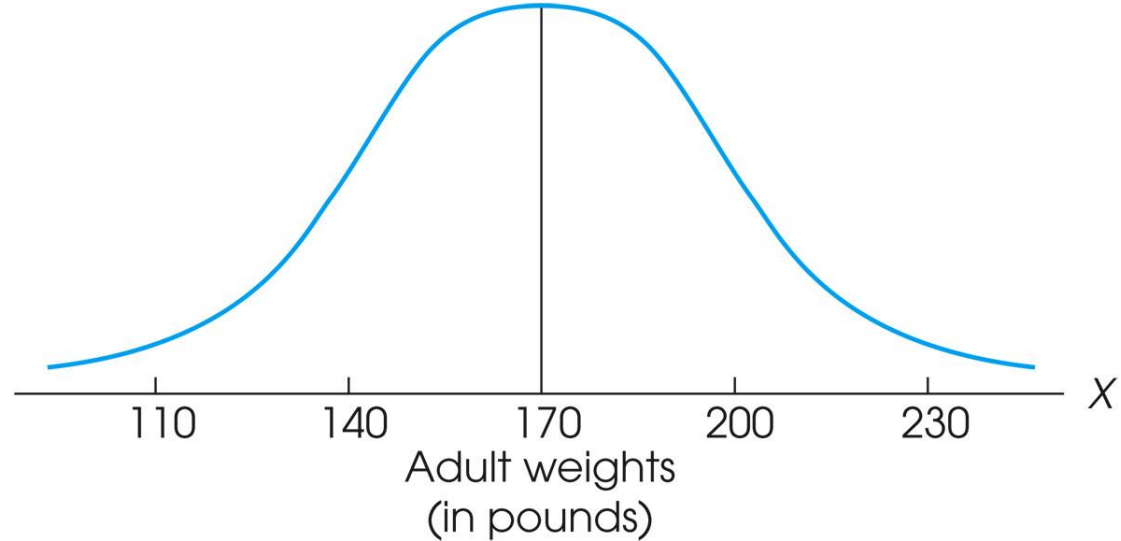
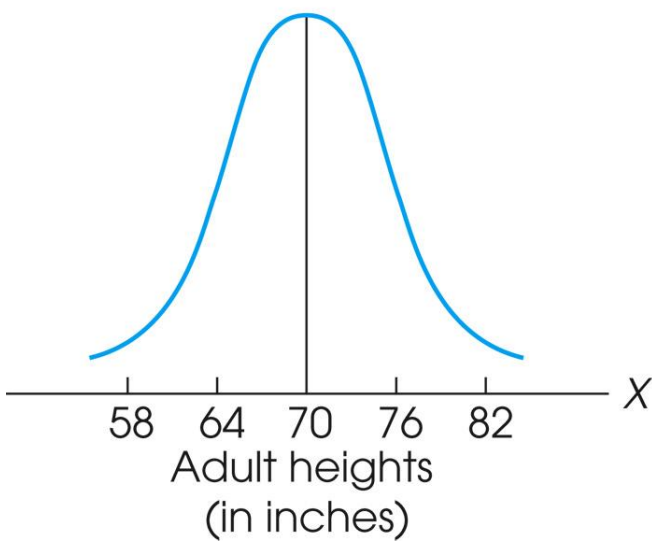
Measures of Variability

4.1 Overview

- Variability can be defined several ways
 - A quantitative distance measure based on the differences between scores
 - Describes distance of the spread of scores or distance of a score from the mean
- Purposes of Measure of Variability
 - Describe the distribution
 - Measure how well an individual score represents the distribution

Figure 4.1

Population Distributions



Three Measures of Variability

- The Range
- The Variance
- The Standard Deviation

4.2 The Range

- The distance covered by the scores in a distribution
 - From smallest value to highest value
- For discrete data, the range is defined as the maximum score minus the minimum score
- For continuous data, real limits are used

$$\text{range} = \text{URL for } X_{\max} - \text{LRL for } X_{\min}$$

- Based on two scores, not all the data
 - An imprecise, unreliable measure of variability

4.3 Standard Deviation and Variance for a Population

- Most common and most important measure of variability is the standard deviation
 - A measure of the standard, or average, distance from the mean
 - Describes whether the scores are clustered closely around the mean or are widely scattered
- Calculation differs for population and samples
- Variance is a necessary ***companion concept*** to standard deviation but ***not*** the same concept

Defining the Standard Deviation

- Step One: Determine the Deviation
- Deviation is distance from the mean

$$\text{Deviation score} = X - \mu$$

- Step Two: Find a “sum of deviations” to use as a basis of finding an “average deviation”
 - Two problems
 - Deviations sum to 0 (because μ is balance point)
 - If sum always 0, “Mean Deviation” will always be 0.
 - Need a new strategy!

Activity for Active Learning

- Group discussion (10-15 minutes)
- Think of an intuitive method for developing a measure of variability based on deviation scores
- Present the idea to the class

Defining the Standard Deviation

(continued)

- Step Two Revised: Remove negative deviations
 - **First** square each deviation score
 - Then sum the Squared Deviations (SS)
- Step Three: Average the squared deviations
 - Mean Squared Deviation is known as “**Variance**”
 - Variability is now measured in squared units

Population variance equals mean (average) squared deviation (distance) of the scores from the population mean

Defining the Standard Deviation

(continued)

- Step Four:
 - Goal: to compute a measure of the “standard” (average) distance of the scores from the mean
 - Variance measures the average squared distance from the mean—not quite our goal
- Adjust for having squared all the differences by taking the square root of the variance
- $\textit{Standard Deviation} = \sqrt{\textit{Variance}}$

Figure 4.2

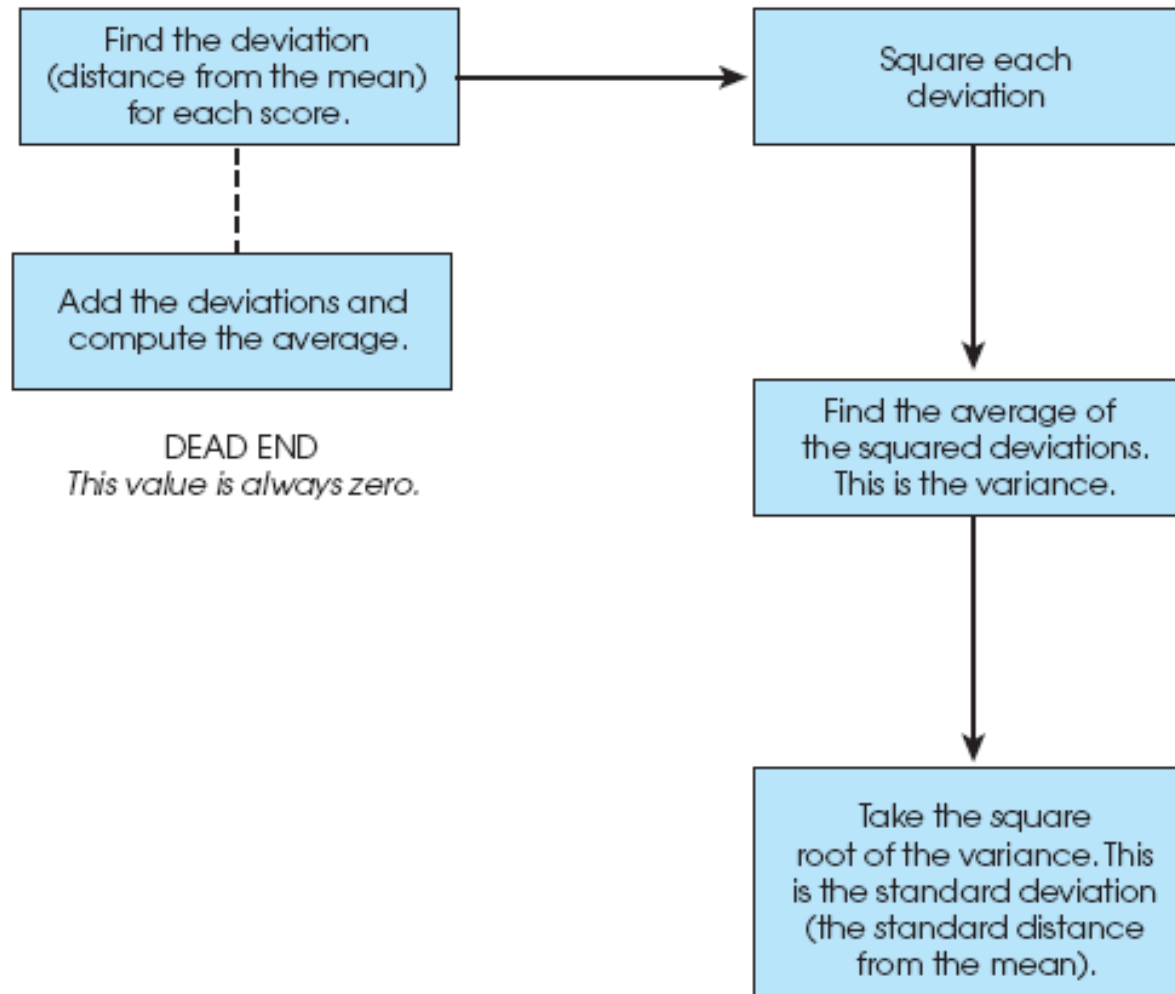
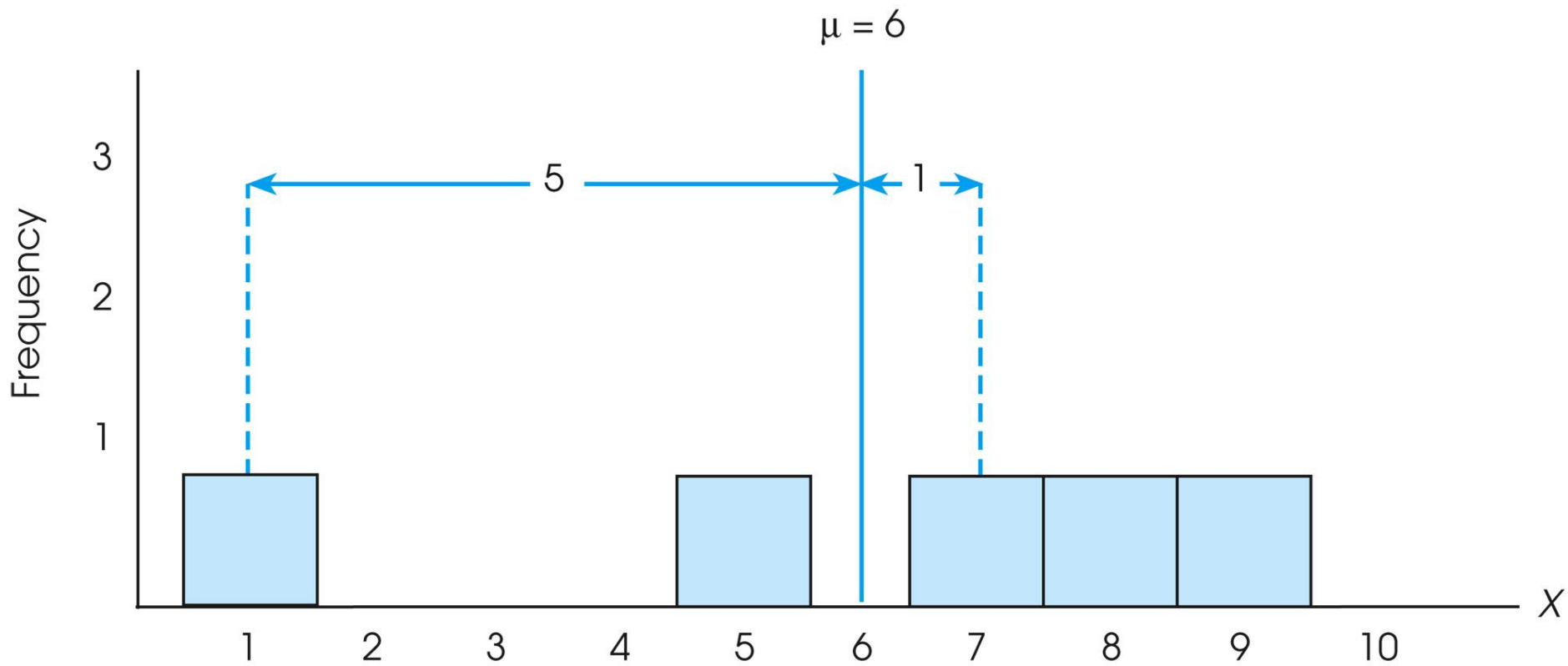


Figure 4.3



Population Variance Formula

$$\text{Variance} = \frac{\text{sum of squared deviations}}{\text{number of scores}}$$

- SS (sum of squares) is the sum of the squared deviations of scores from the mean
- Two formulas for computing SS

Two formulas for SS

Definitional Formula

- Find each deviation score $(X - \mu)$
- Square each deviation score, $(X - \mu)^2$
- Sum up the squared deviations

$$SS = \sum (X - \mu)^2$$

Computational Formula

- Square each score and sum the squared scores
- Find the sum of scores, square it, divide by N
- Subtract the second part from the first

$$SS = \sum X^2 - \frac{(\sum X)^2}{N}$$

Caution Required!

When using the computational formula, remember...

$$SS \neq \sum X^2$$

$$SS \neq (\sum X)^2$$

$$\sum X^2 \neq (\sum X)^2$$

Population Variance: Formula and Notation

Formula

$$\text{variance} = \frac{SS}{N}$$

$$\text{standard deviation} = \sqrt{\frac{SS}{N}}$$

Notation

- Variance is the average of squared deviations, so we identify population variance with a lowercase Greek letter sigma squared: σ^2
- Standard deviation is the square root of the variance, so we identify it with a lowercase Greek letter sigma: σ

Learning Check

- Decide if each of the following statements is True or False.

T/F

- The computational & definitional formulas for SS sometimes give different results

T/F

- If all the scores in a data set are the same, the Standard Deviation is equal to 1.00

Learning Check - Answer

False

- The computational formula is just an algebraic rearrangement of the definitional formula. Results are identical

False

- When all the scores are the same, they are all equal to the mean. Their deviations = 0, as does their Standard Deviation

Learning Check

- The standard deviation measures ...

A

- Sum of squared deviation scores

B

- Standard distance of a score from the mean

C

- Average deviation of a score from the mean

D

- Average squared distance of a score from the mean

Learning Check - Answer

- The standard deviation measures ...

A

- Sum of squared deviation scores

B

- Standard distance of a score from the mean

C

- Average deviation of a score from the mean

D

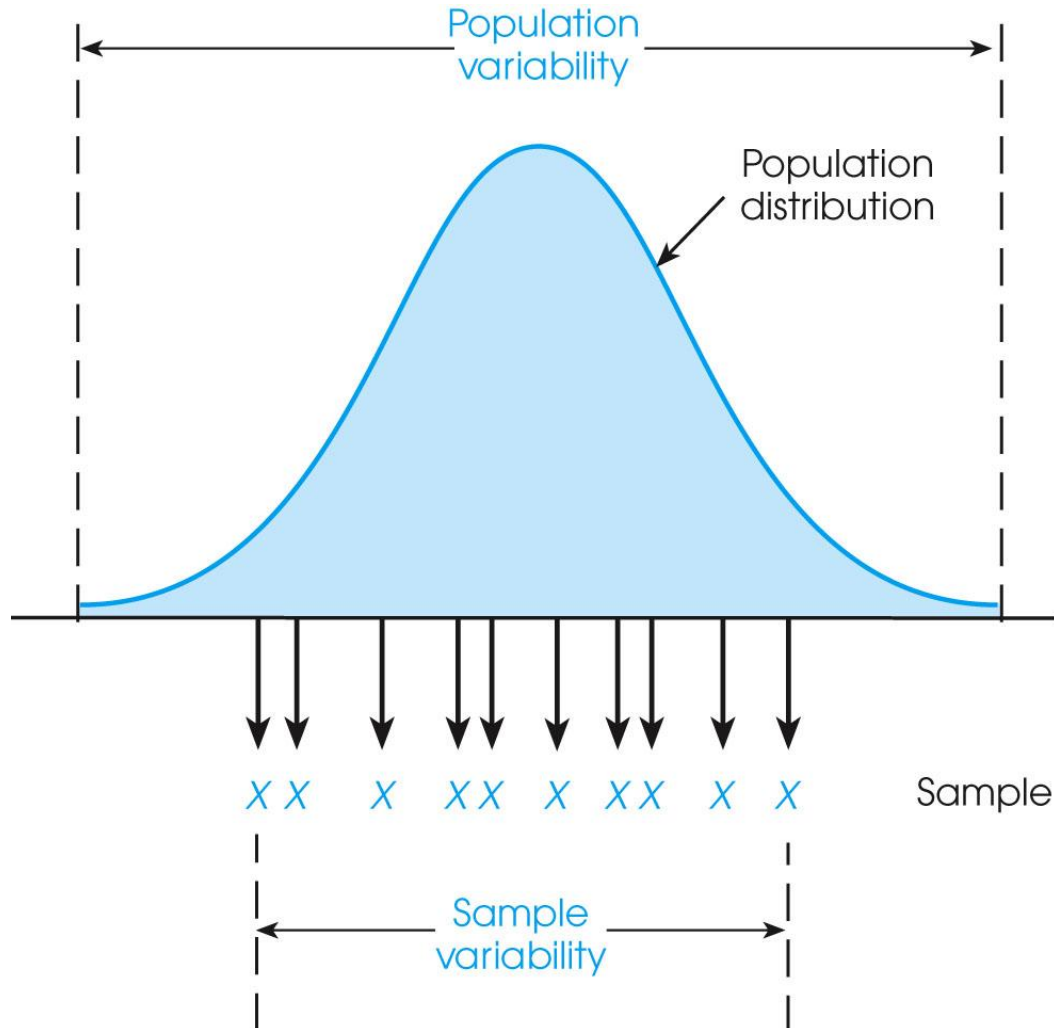
- Average squared distance of a score from the mean

4.4 Standard Deviation and Variance for a Sample

- Goal of inferential statistics:
 - Draw general conclusions about population
 - Based on limited information from a sample
- Samples differ from the population
 - Samples have less variability
 - Computing the Variance and Standard Deviation in the same way as for a population would give a biased estimate of the population values

Figure 4.4

Population of Adult Heights



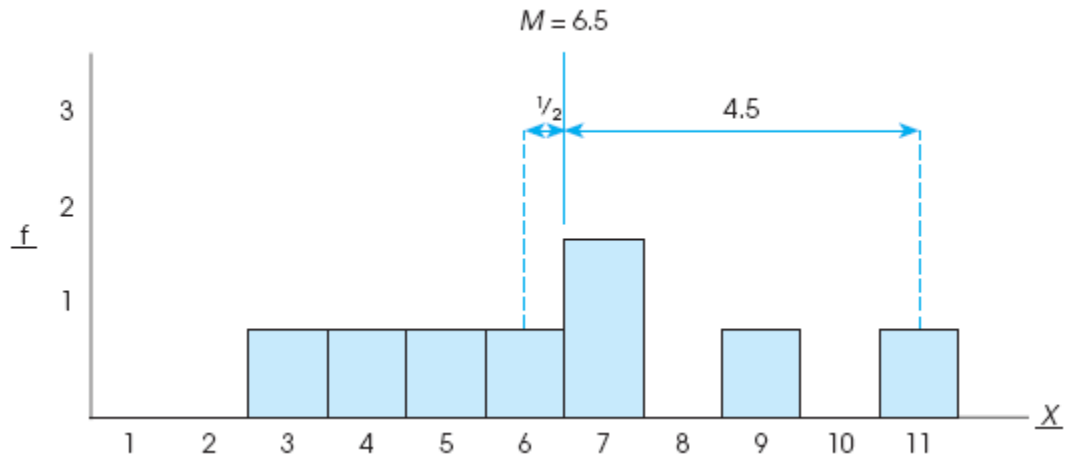
Sample Variance and Standard Deviation

- Sum of Squares (SS) is computed as before
- Formula for Variance has $n-1$ rather than N in the denominator
- Notation uses s instead of σ

$$\text{variance of sample} = s^2 = \frac{SS}{n-1}$$

$$\text{standard deviation of sample} = s = \sqrt{\frac{SS}{n-1}}$$

Figure 4.5
Histogram for Sample of $n = 8$



Degrees of Freedom

- Population variance
 - Mean is known
 - Deviations are computed from a known mean
- Sample variance as estimate of population
 - Population mean is unknown
 - Using sample mean restricts variability
- Degrees of freedom
 - Number of scores in sample that are independent and free to vary
 - Degrees of freedom (df) = $n - 1$

Learning Check

- A sample of four scores has $SS = 24$.
What is the variance?

A

- The variance is 6

B

- The variance is 7

C

- The variance is 8

D

- The variance is 12

Learning Check - Answer

- A sample of four scores has $SS = 24$.
What is the variance?

A

- The variance is 6

B

- The variance is 7

C

- **The variance is 8**

D

- The variance is 12

Learning Check

- Decide if each of the following statements is True or False.

T/F

- A sample systematically has less variability than a population

T/F

- The standard deviation is the distance from the Mean to the farthest point on the distribution curve

Learning Check - Answer

True

- Extreme scores affect variability, but are less likely to be included in a sample

False

- The standard deviation extends from the mean approximately halfway to the most extreme score

4.5 More About Variance and Standard Deviation

- Mean and standard deviation are particularly useful in clarifying graphs of distributions
- Means and standard deviations together provide extremely useful descriptive statistics for characterizing distributions

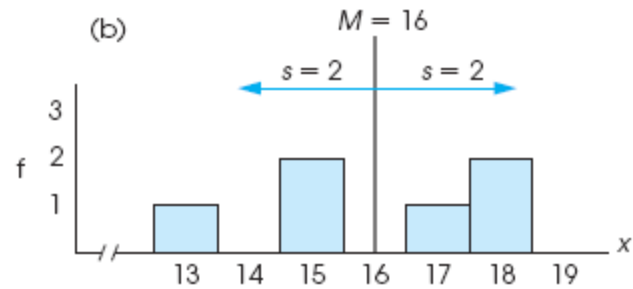
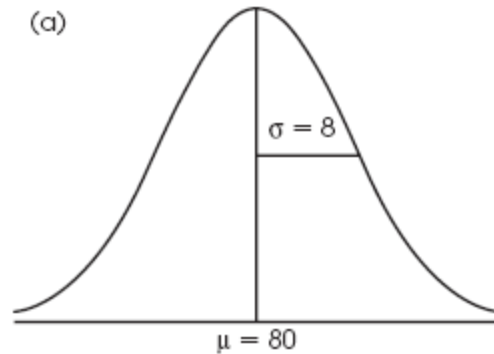
Biased and unbiased statistics

- A sample statistic is unbiased if the average value of the statistic is equal to the population parameter. (The average value of the statistic is obtained from all the possible samples for a specific sample size, n .)
- A sample statistic is biased if the average value of the statistic either underestimates or overestimates the corresponding population parameter.

Showing Mean and Standard Deviation in a Graph

- For both populations and samples it is easy to represent mean and standard deviation
 - Vertical line in the “center” denotes location of mean
 - Horizontal line to right, left (or both) denotes the distance of one standard deviation

Figure 4.6 Showing Means and Standard Deviations in Graphs



Sample Variance as an Unbiased Statistic

- Unbiased estimate of a population parameter
 - Average value of statistic is equal to parameter
 - Average value uses all possible samples of a particular size n
 - Corrected standard deviation formula (dividing by $n-1$) produces an *unbiased estimate* of the population variance
- Biased estimate of a population parameter
 - Systematically overestimates or underestimates the population parameter

Example

- Consider a population that consists of exactly $N = 6$ scores: 0, 0, 3, 3, 9, 9
- This population has a mean of 4
- And a variance of 14
- Now, select samples of $n = 2$ scores from this population
- The complete set of samples is listed on Table 4.1

Table 4.1 Biased and Unbiased Estimates

			Sample Statistics		
Sample	1 st Score	2 nd Score	(Unbiased) Mean	Biased Variance	Unbiased Variance
1	0	0	0.00	0.00	0.00
2	0	3	1.50	2.25	4.50
3	0	9	4.50	20.25	40.50
4	3	0	1.50	2.25	4.50
5	3	3	3.00	0.00	0.00
6	3	9	6.00	9.00	18.00
7	9	0	4.50	20.25	40.50
8	9	3	6.00	9.00	18.00
9	9	9	9.00	0.00	0.00
Totals			36.00	63.00	126.00

Mean of 9 unbiased sample mean estimates: $36/9 = 4$ (Actual $\mu = 4$)

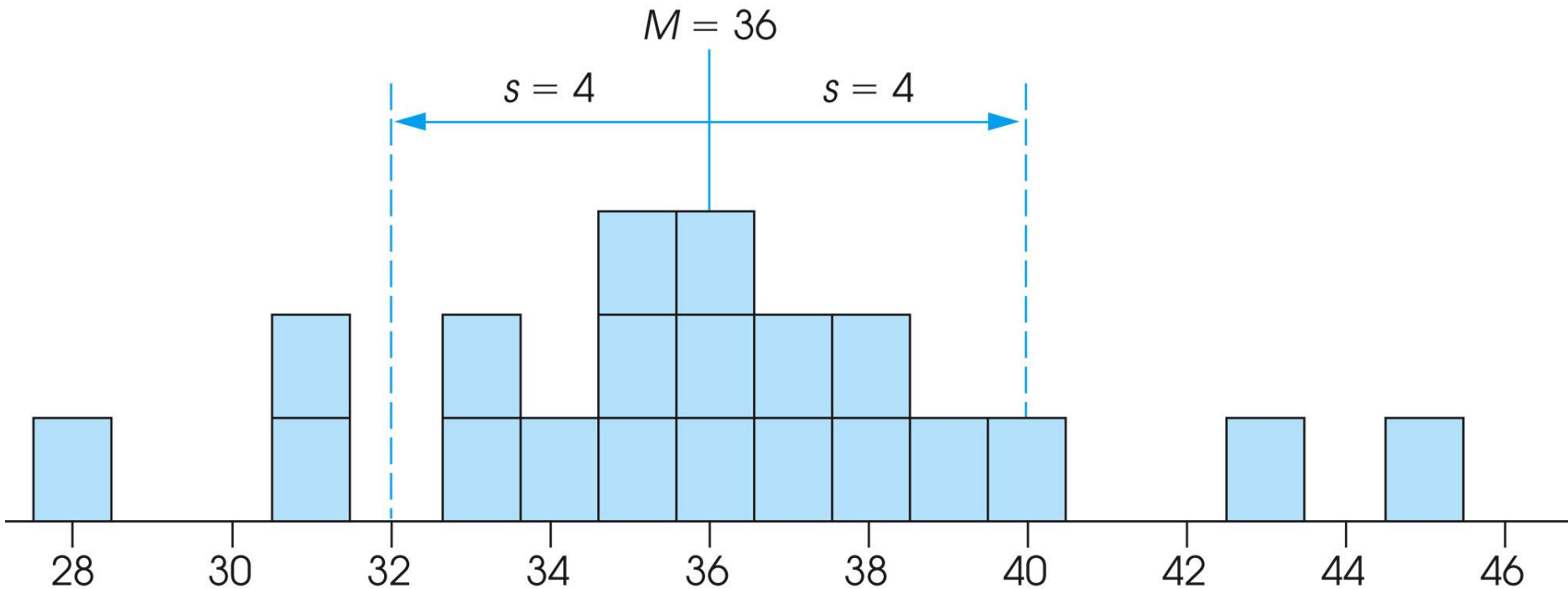
Mean of 9 biased sample variance estimates: $63/9 = 7$ (Actual $\sigma^2 = 14$)

Mean of 9 unbiased sample variance estimates: $126/9 = 14$ (Actual $\sigma^2 = 14$)

Standard Deviation and Descriptive Statistics

- A standard deviation describes scores in terms of *distance from the mean*
- Describe an entire distribution with just two numbers (M and s)
- Reference to both allows reconstruction of the measurement scale from just these two numbers (Figure 4.7)

Figure 4.7
(Sample) $n = 20$, $M = 36$, and $s = 4$



Transformations of Scale

- Adding a constant to each score
 - The Mean is changed
 - The standard deviation is unchanged
- Multiplying each score by a constant
 - The Mean is changed
 - Standard Deviation is also changed
 - The Standard Deviation is multiplied by that constant

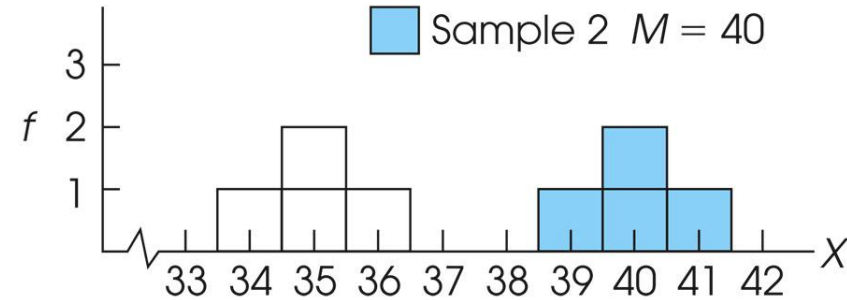
Variance and Inferential Statistics

- Goal of inferential statistics is to detect meaningful and significant patterns in research results
- Variability in the data influences how easy it is to see patterns
 - High variability obscures patterns that would be visible in low variability samples
 - Variability is sometimes called error variance

Figure 4.8 Experiments with high and low variability

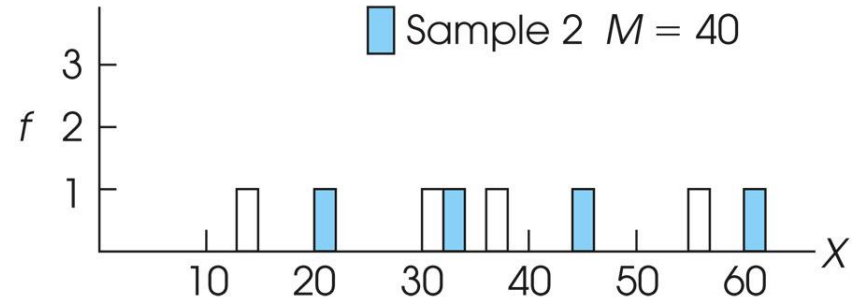
Data from Experiment A

□ Sample 1 $M = 35$
■ Sample 2 $M = 40$



Data from Experiment B

□ Sample 1 $M = 35$
■ Sample 2 $M = 40$



Learning Check

- A population has $\mu = 6$ and $\sigma = 2$. Each score is multiplied by 10. What is the shape of the resulting distribution?

A

- $\mu = 60$ and $\sigma = 2$

B

- $\mu = 6$ and $\sigma = 20$

C

- $\mu = 60$ and $\sigma = 20$

D

- $\mu = 6$ and $\sigma = 5$

Learning Check - Answer

- A population has $\mu = 6$ and $\sigma = 2$. Each score is multiplied by 10. What is the shape of the resulting distribution?

A

• $\mu = 60$ and $\sigma = 2$

B

• $\mu = 6$ and $\sigma = 20$

C

• $\mu = 60$ and $\sigma = 20$

D

• $\mu = 6$ and $\sigma = 5$

Learning Check TF

- Decide if each of the following statements is True or False.

T/F

- A biased statistic has been influenced by researcher error

T/F

- On average, an unbiased sample statistic has the same value as the population parameter

Learning Check - Answer

False

- Bias refers to the systematic effect of using sample data to estimate a population parameter

True

- Each sample's statistic differs from the population parameter, but the average of all samples will equal the parameter

**Figure 4.9 SPSS Summary Table ($n = 8$)
from Example 4.5**

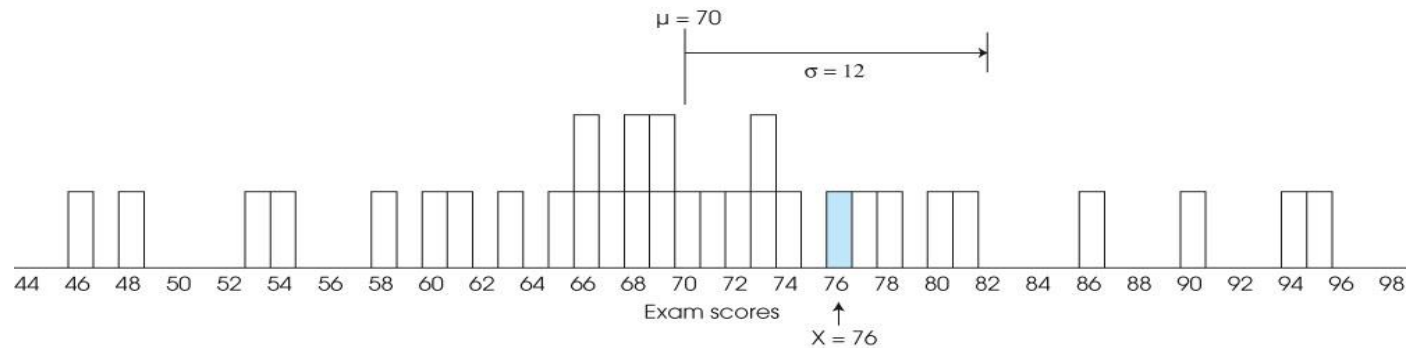
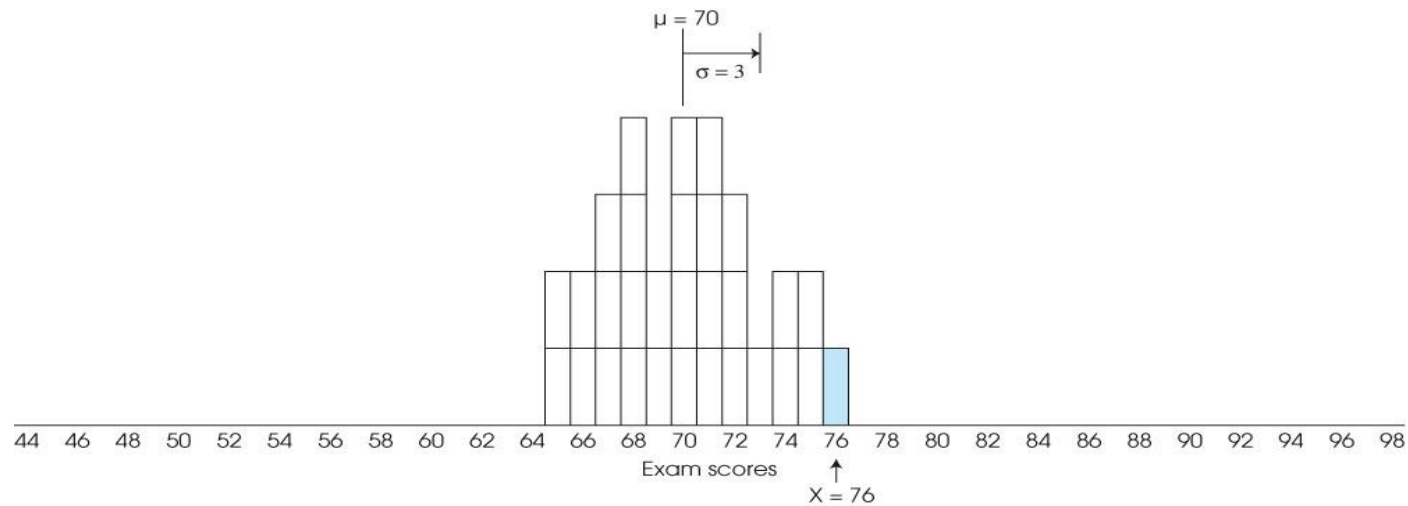
	N	Range	Minimum	Maximum	Mean	Std. Deviation	Variance
VAR00001 Valid N (listwise)	8 8	8.00	3.00	11.00	6.5000	2.61861	6.85714

Chapter 5 z-Scores

5.1 Purpose of z-Scores

- Identify and describe exact location of individual scores in a distribution.
- Standardize an entire distribution
 - Takes different distributions and makes them equivalent and comparable

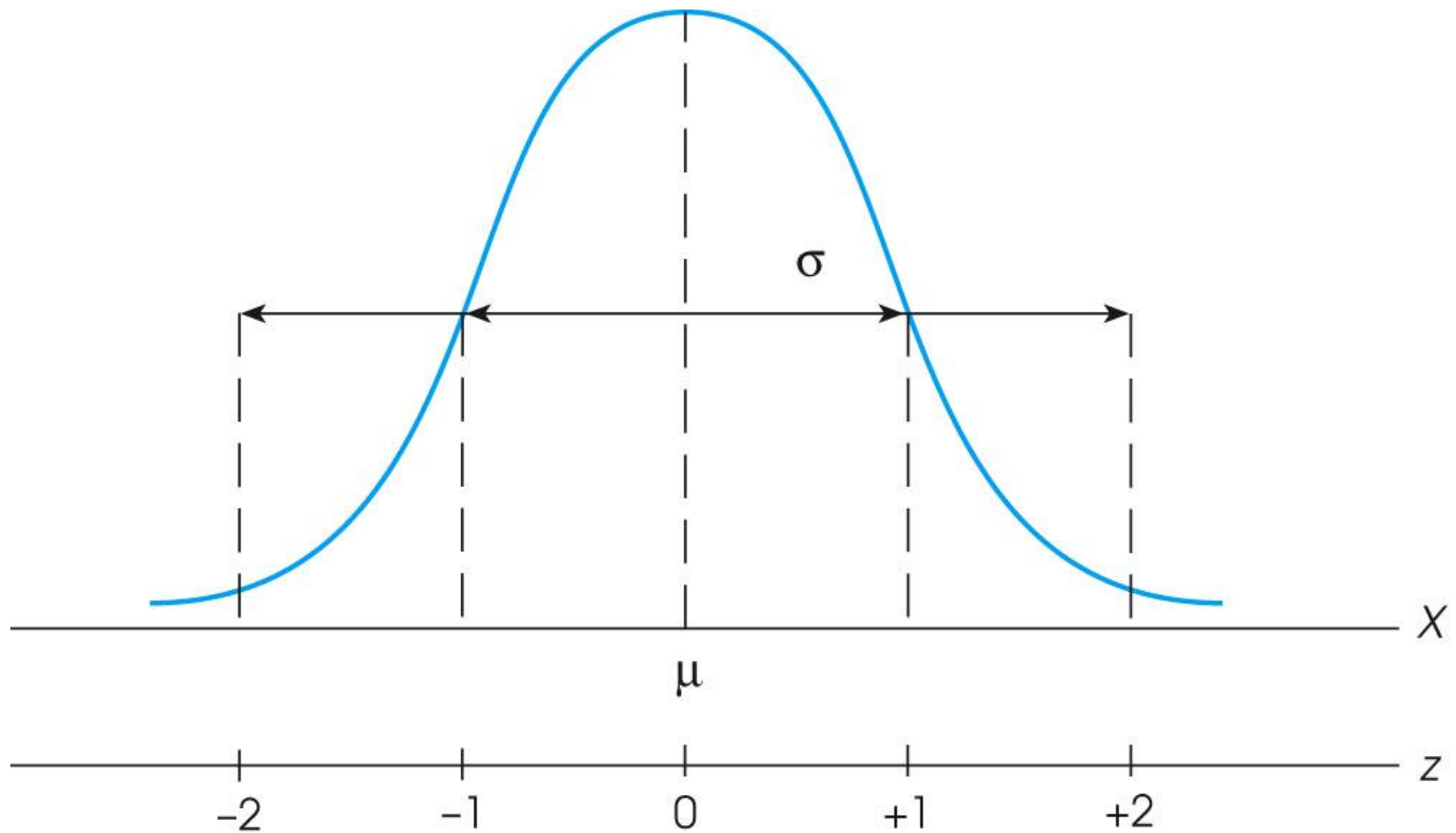
Figure 5.1 Two distributions of exam scores



5.2 Locations and Distributions

- Exact location of a raw score (X value) is described by z-score:
 - **Sign** (+ or -) tells whether score is located above or below the mean
 - **Number** tells distance between score and mean in standard deviation units

Figure 5.2 Relationship of z-scores and locations



Learning Check

- A z-score of $z = +1.00$ indicates a position in a distribution _____

A

- Above the mean by 1 point

B

- Above the mean by a distance equal to 1 standard deviation

C

- Below the mean by 1 point

D

- Below the mean by a distance equal to 1 standard deviation

Learning Check - Answer

- A z-score of $z = +1.00$ indicates a position in a distribution _____

A

- Above the mean by 1 point

B

- **Above the mean by a distance equal to 1 standard deviation**

C

- Below the mean by 1 point

D

- Below the mean by a distance equal to 1 standard deviation

Learning Check

- Decide if each of the following statements is **True** or **False**.

T/F

- A negative z-score always indicates a location below the mean

T/F

- A score close to the mean has a z-score close to 1.00

Answer

True

- Sign indicates that score is below the mean

False

- Scores close to the mean have z-scores close to 0.00

Equation for z-score

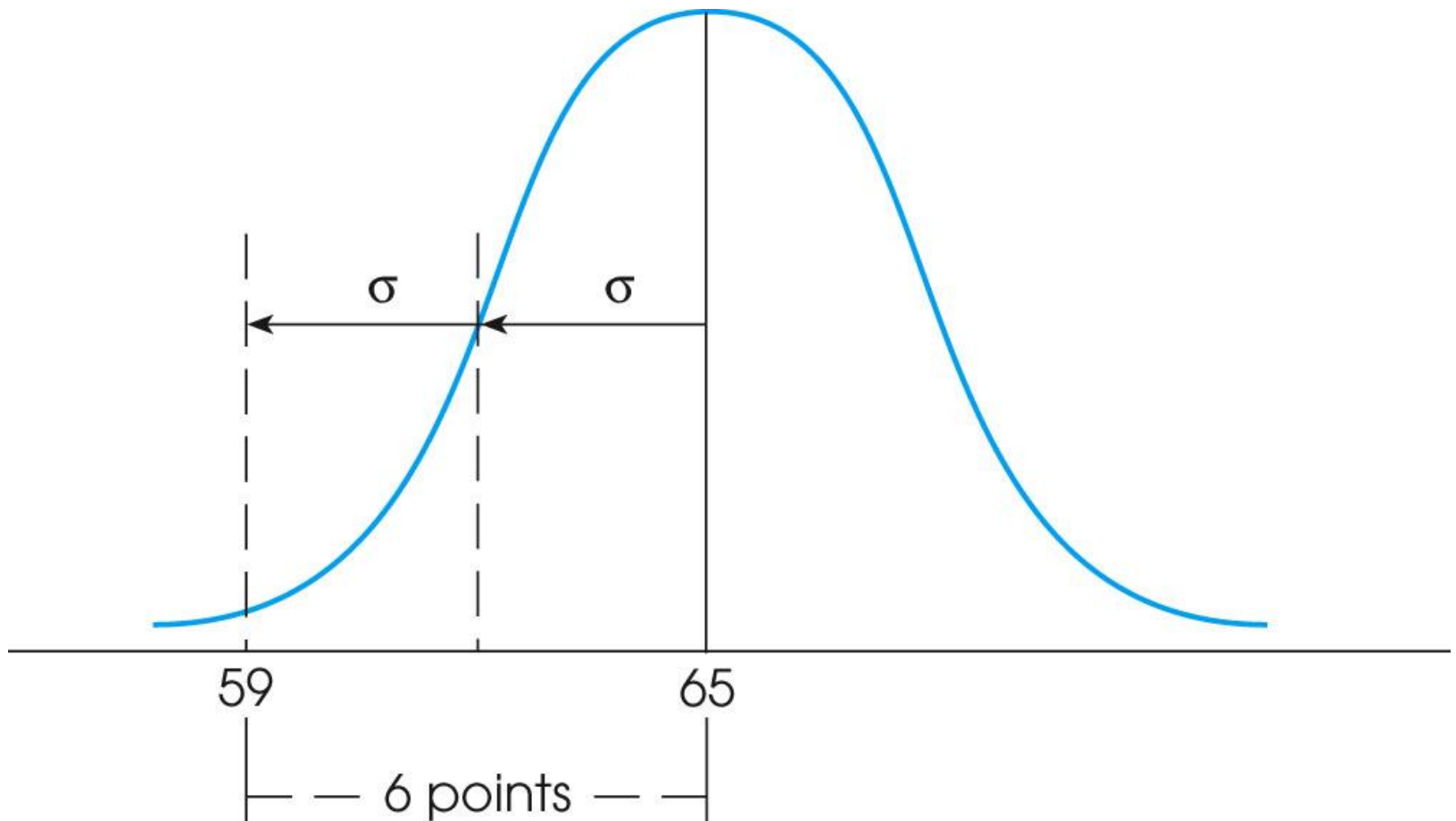
$$z = \frac{X - \mu}{\sigma}$$

- Numerator is a **deviation score**
- Denominator expresses deviation in standard deviation units

Determining raw score from z-score

$$z = \frac{X - \mu}{\sigma} \quad \text{so} \quad X = \mu + z\sigma$$

Figure 5.3 Example 5.4



Learning Check

- For a population with $\mu = 50$ and $\sigma = 10$, what is the X value corresponding to $z=0.4$?

A

• 50.4

B

• 10

C

• 54

D

• 10.4

Learning Check - Answer

- For a population with $\mu = 50$ and $\sigma = 10$, what is the X value corresponding to $z=0.4$?

A

• 50.4

B

• 10

C

• 54

D

• 10.4

Learning Check

- Decide if each of the following statements is **True** or **False**.

T/F

- If $\mu = 40$ and $X = 50$ corresponds to $z = +2.00$, then $\sigma = 5$ points

T/F

- If $\sigma = 20$, a score above the mean by 10 points will have $z = 1.00$

Answer

True

- If $2\sigma = 10$ then $\sigma = 5$

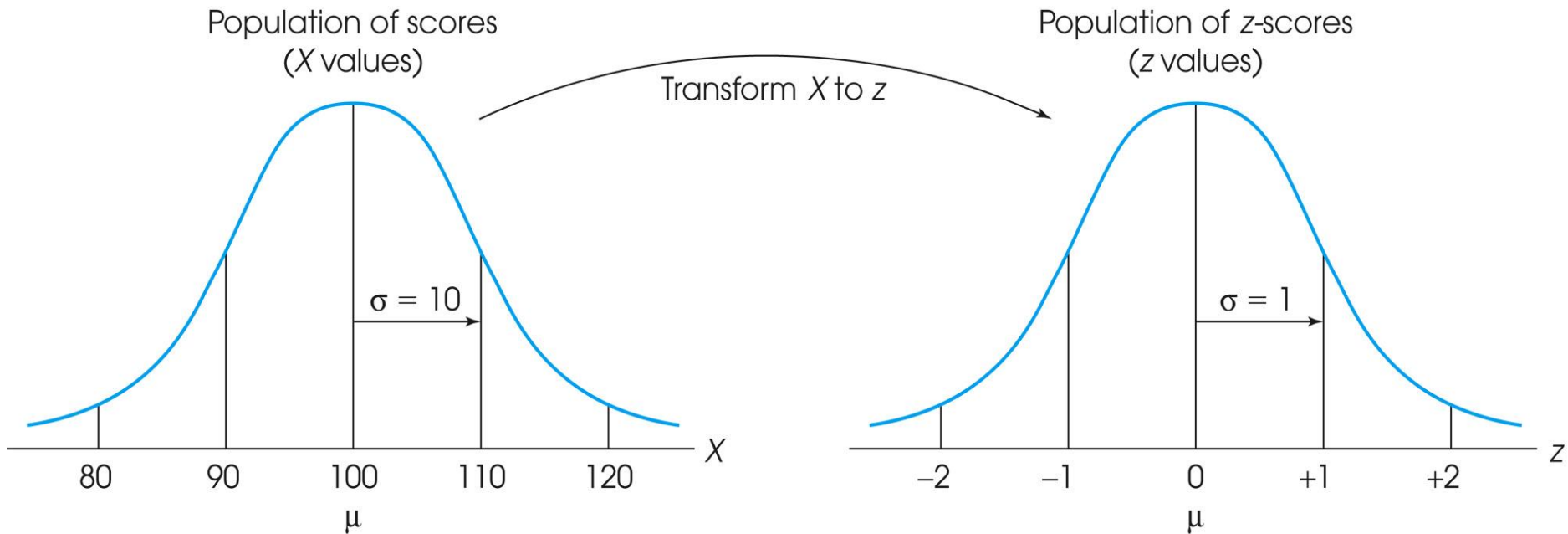
False

- Why? A score above the mean by 10 points would be above the mean by 0.5 standard deviations ($10/20$). So that $z = 0.5$.

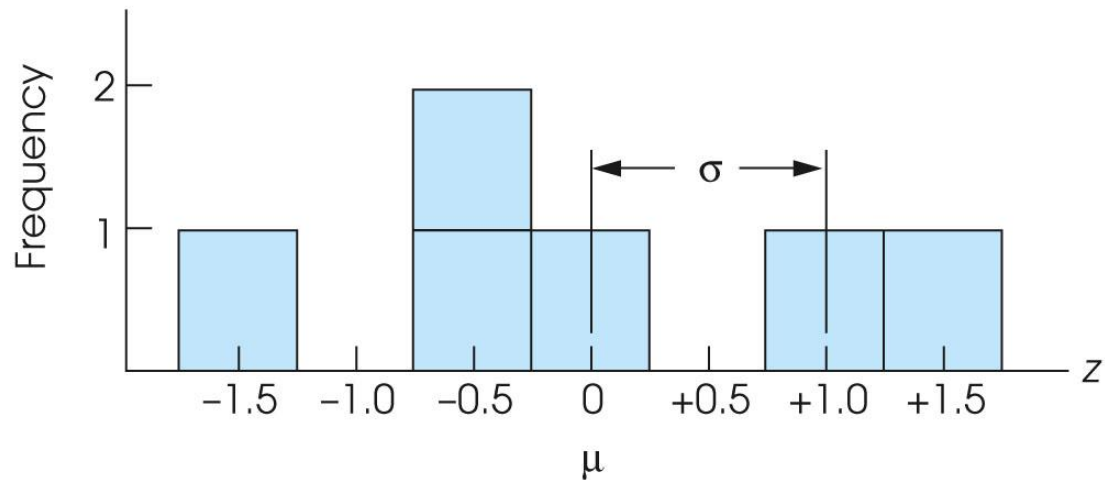
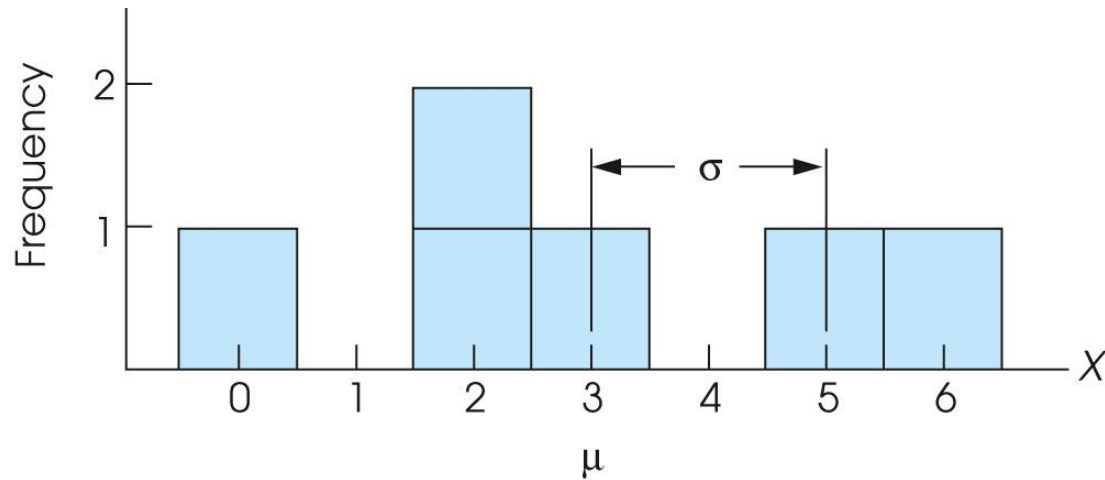
5.3 Standardizing a Distribution

- Every X value can be transformed to a z-score
- Characteristics of **z-score distribution after transformation**:
 - Same **shape** as original distribution
 - **Mean** of a z-score distribution is **always 0**.
 - **Standard deviation** is **always 1.00**
- A z-score distribution is called a **standardized distribution**

Figure 5.4 Transformation of a Population of Scores



Shape of Distribution after z-Score Transformation Stays the Same



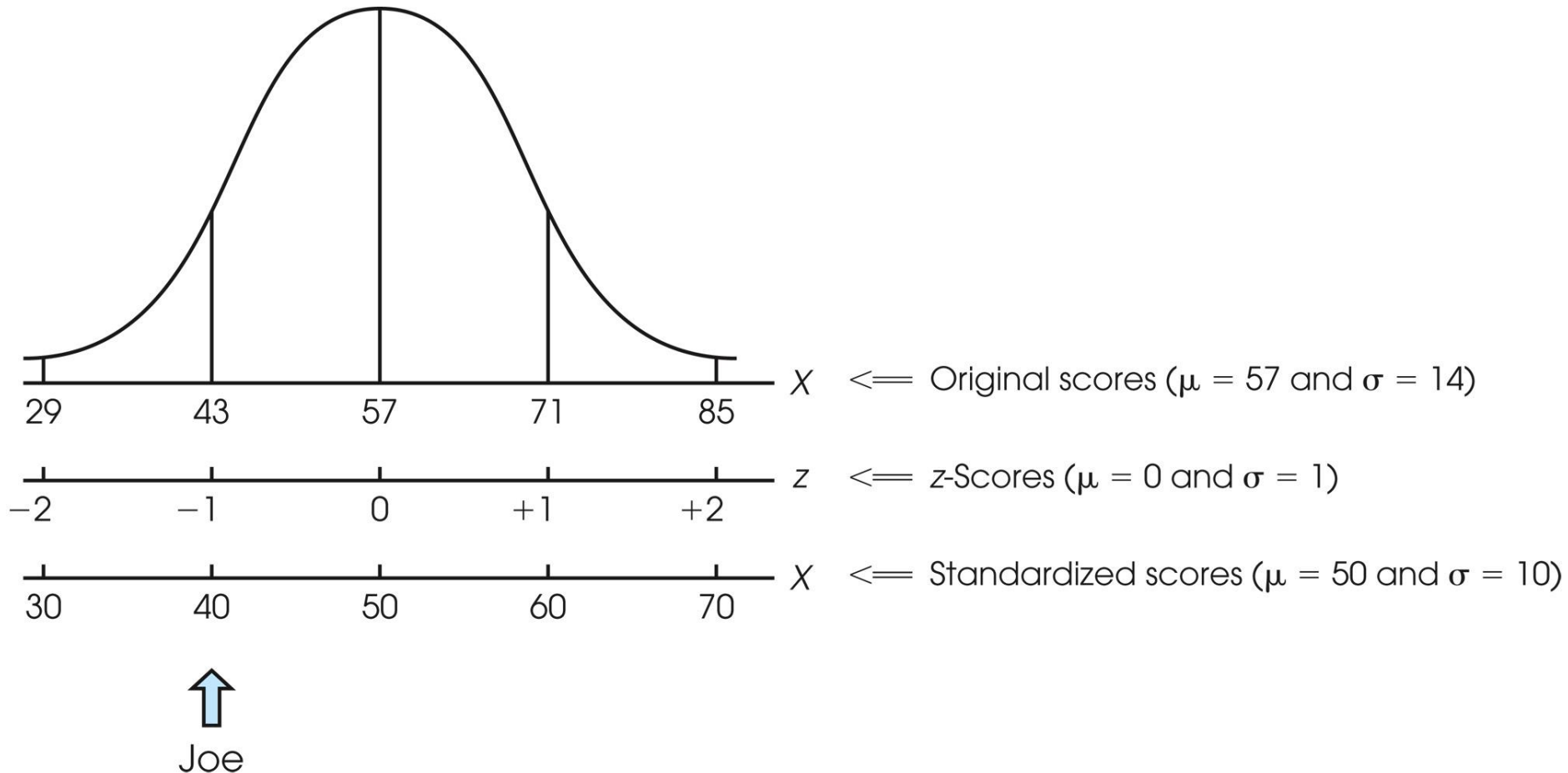
z-Scores for Comparisons

- Scores from different distributions can be converted to z-scores
- The z-scores (standardized scores) allow the comparison of scores from two different distributions because all z-scores are comparable to each other

5.4 Other Standardized Distributions

- Process of standardization is widely used
 - SAT has $\mu = 500$ and $\sigma = 100$
 - IQ has $\mu = 100$ and $\sigma = 15$ Point
- Standardizing a distribution has two steps
 - Raw scores (X values) transformed to z-scores
 - Z-scores transformed to new X values so that the specific μ and σ are attained.

Figure 5.7 Creating a Standardized Distribution



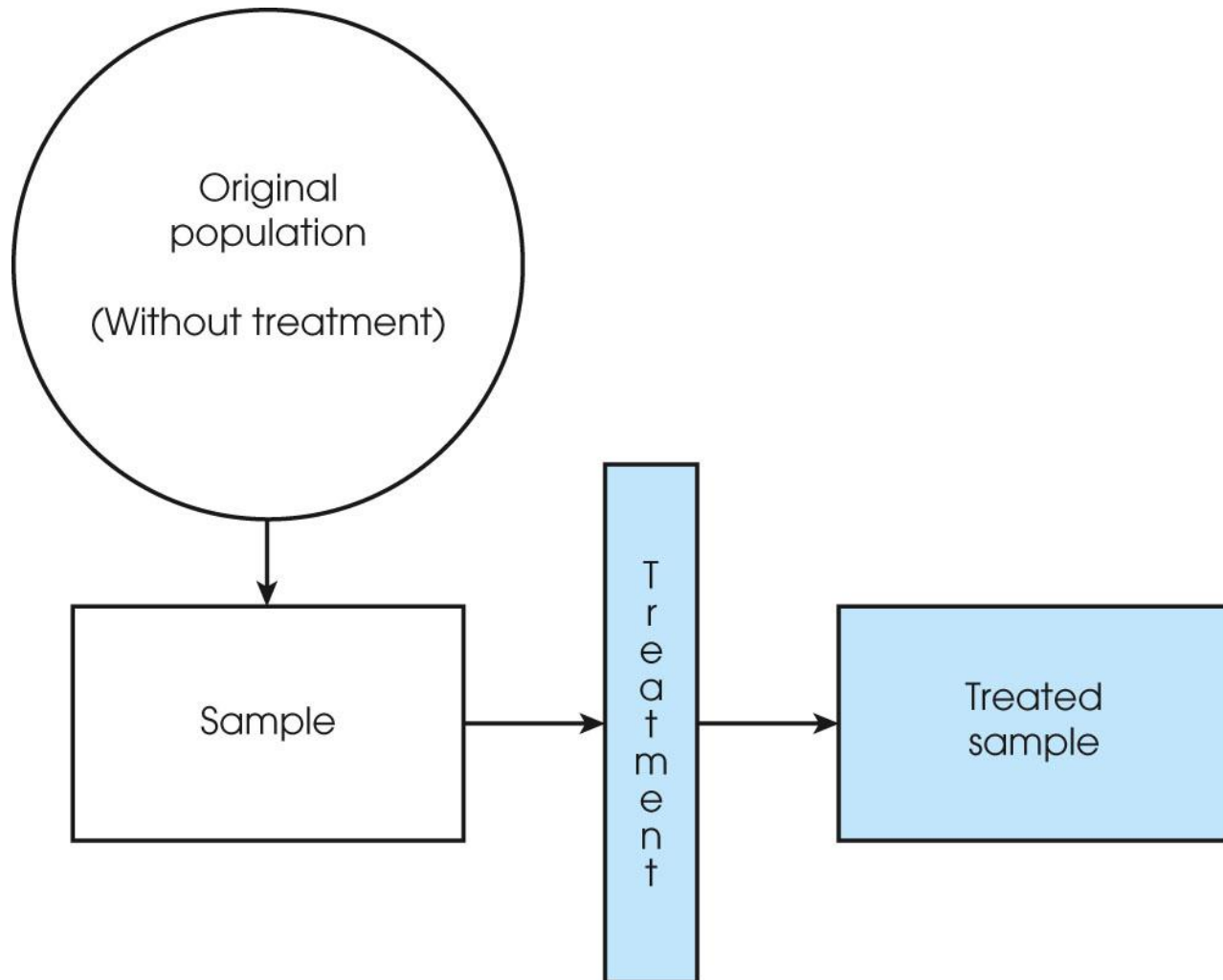
5.5 Computing z-Scores for Samples

- Populations are most common context for computing z-scores
- It is possible to compute z-scores for samples
 - Indicates relative position of score in sample
 - Indicates distance from sample mean
- Sample distribution can be transformed into z-scores
 - Same shape as original distribution
 - $M = 0$ and $s = 1$

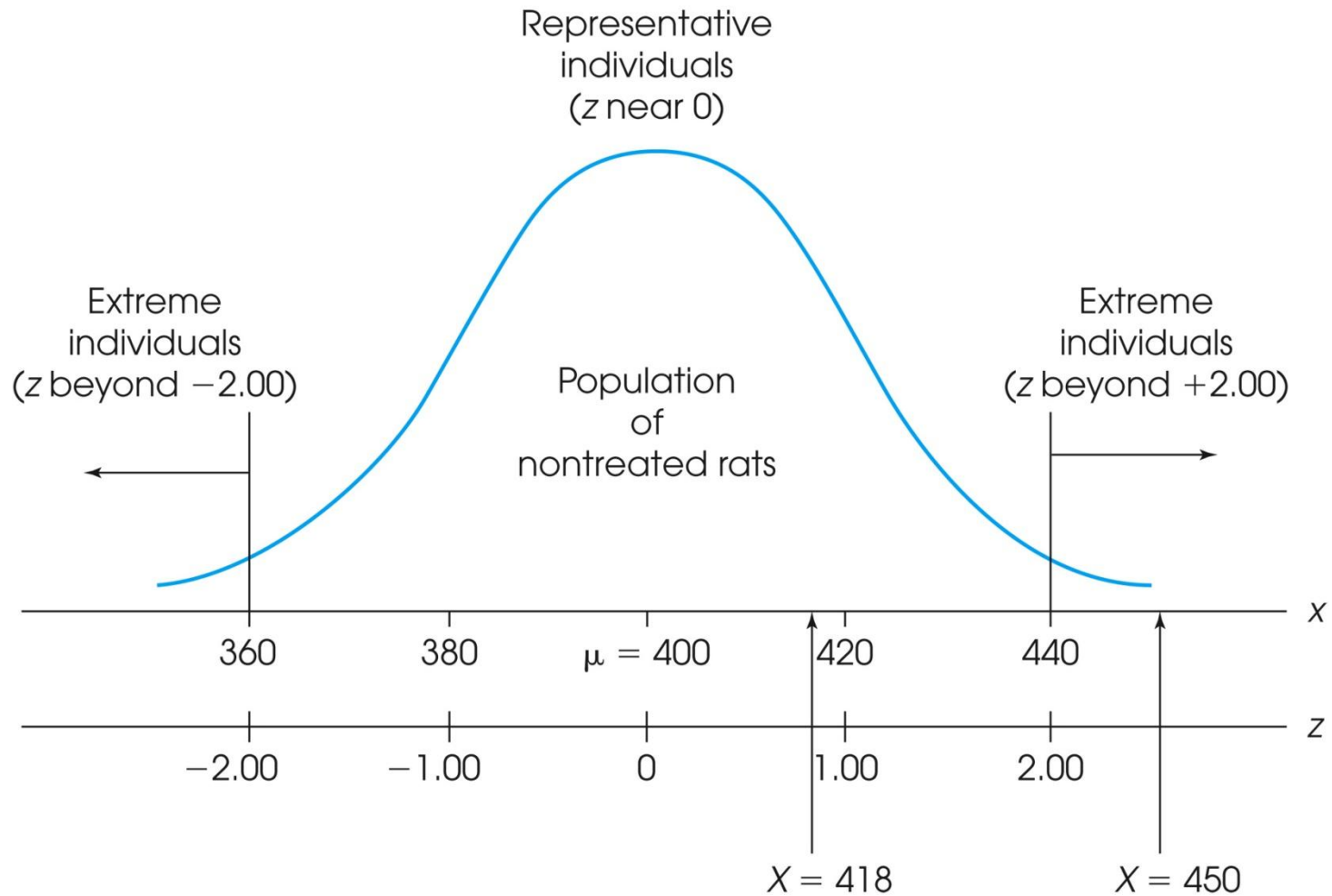
5.6 Looking to Inferential Statistics

- Interpretation of research results depends on determining if (treated) sample is *noticeably different* from the population
- One technique for defining *noticeably different* uses z-scores.

Diagram of Research Study



Distributions of weights



Learning Check

- Last week Andi had exams in Chemistry and in Spanish. On the chemistry exam, the mean was $\mu = 30$ with $\sigma = 5$, and Andi had a score of $X = 45$. On the Spanish exam, the mean was $\mu = 60$ with $\sigma = 6$ and Andi had a score of $X = 65$. For which class should Andi expect the better grade?

A

• **Chemistry**

B

• **Spanish**

C

• **There is not enough information to know**

Learning Check - Answer

- Last week Andi had exams in Chemistry and in Spanish. On the chemistry exam, the mean was $\mu = 30$ with $\sigma = 5$, and Andi had a score of $X = 45$. On the Spanish exam, the mean was $\mu = 60$ with $\sigma = 6$ and Andi had a score of $X = 65$. For which class should Andi expect the better grade?

A

• **Chemistry**

B

• Spanish

C

• There is not enough information to know

Homework

- Chapter 4:
 - Q 6
 - Q 20
 - Q 22
- Chapter 5:
 - Q 6
 - Q 8
 - Q 12
 - Q 24